

Khaula Molapo

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Artificial Intelligence

Week: 7

1. Scenario Analysis

An expert system can be used by a bank to assist with loan approval. The system stores knowledge about loan requirements and uses rules to evaluate information provided by an applicant. For example, the system can consider the applicant's income, employment status, credit history, existing debts, and requested loan amount. If an applicant has a stable income, a good credit history, and manageable debt, the system may determine that the application meets the required conditions. If the applicant has a poor credit history or excessive debt, the system can identify the application as high risk and refer it for further assessment. The expert system does not simply store information; it applies predefined rules to the available facts and produces a conclusion. A human loan officer can then review the result before making the final decision. This type of system can improve consistency, reduce repetitive work, and help process large numbers of applications more efficiently.

2. Concept Research

CLIPS is an expert-system shell developed for building rule-based artificial intelligence applications. It provides the tools needed to represent facts, create rules, and perform reasoning using those rules. Instead of requiring developers to build the entire reasoning system from scratch, CLIPS provides an inference engine that determines which rules should be activated when certain facts are available. A typical CLIPS system contains facts describing a situation and rules that determine what actions or conclusions should follow. It can be used for applications such as diagnosis, decision support, and problem solving. CLIPS is useful for learning expert systems because its rule-based structure clearly demonstrates how knowledge and inference work together. It is also designed to handle changing facts while a system is running.

CLIPS implementation:

```
(deffacts computer-problems
  (no-internet)
  (wifi-disabled)
  (low-battery)
  (low-storage)
  (overheating))
```

```

(defrule internet-problem
  (no-internet)
  =>
  (printout t "Check the network connection." crlf))

(defrule wifi-problem
  (wifi-disabled)
  =>
  (printout t "Enable Wi-Fi." crlf))

(defrule battery-problem
  (low-battery)
  =>
  (printout t "Connect the computer to a power source." crlf))

(defrule storage-problem
  (low-storage)
  =>
  (printout t "Remove unnecessary files." crlf))

(defrule overheating-problem
  (overheating)
  =>
  (printout t "Allow the computer to cool down and check ventilation."
  crlf))

```

3. Tool Practice

For the tool practice, I created a simple five-rule expert system using CLIPS. The system uses facts about a computer problem and applies rules to identify a possible solution. For example, one rule checks whether a computer has no internet connection, while another checks whether the Wi-Fi is disabled. Additional rules can identify problems such as low battery, insufficient storage, or an overheated computer. I observed that the system does not make decisions randomly. Instead, it checks the available facts and activates rules whose conditions are satisfied. This helped me understand how an expert system separates knowledge from the reasoning process. The activity also showed me that adding more rules can allow the system to handle more situations. However, the system is limited by the quality and completeness of the rules provided by the developer.

4. Diagram Design

The architecture diagram shows the main components of a rule-based expert system and how they interact. The system begins with facts or information provided by a user. These facts are passed to the knowledge base, which contains rules describing how different conditions should be handled. The inference engine compares the available facts with the rules and determines which rules should be activated. The selected rules produce

conclusions or recommendations, which are then presented to the user through the user interface. The diagram can also show a feedback path where new information is entered and the reasoning process is repeated. This structure demonstrates the relationship between the knowledge base and inference engine. It also shows that an expert system depends on both accurate knowledge and an effective reasoning process to produce useful results.